

# Shinu Rathod

Data Scientist · AI Researcher · ML Engineer

AI Researcher at IIT Ropar — Cybersecurity AI Division · Rupnagar, Punjab, India

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## TECHNICAL SKILLS

### Programming

Python   SQL   Kotlin   R

### ML / DL Frameworks

PyTorch   TensorFlow   Keras  
Scikit-learn   Hugging Face  
LoRA / QLoRA   ONNX Runtime

### NLP & Computer Vision

NLP   Transformers   Computer Vision  
Text Classification

### Generative AI (GenAI)

LLM Fine-Tuning   Prompt Engineering  
RAG Pipelines   LangChain   LlamaIndex  
Pinecone   ChromaDB

### Agentic AI

AI Agents   ReAct Framework   Tool Use  
Multi-Agent Systems

### MLOps & Deployment

Flask   FastAPI   Docker   Streamlit  
Android / ONNX

### Visualization

Power BI   Matplotlib   Seaborn

## EDUCATION

### Bachelor of Computer Applications (BCA)

Karnataka College, Bidar, Karnataka  
2018 – 2021  
CGPA: 7.64 / 10

### Data Science Professional Program

360DigiTMG, Hyderabad  
Feb 2023 – Feb 2024

## CERTIFICATIONS

### Data Science & Analytics Internship

Zidio Development · Dec 2024

### Autism Detection Internship

AI SPRY · May 2025

## PROFESSIONAL SUMMARY

AI Researcher at **Indian Institutes of Technology-Ropar(IIT Ropar)** and Data Scientist with hands-on expertise in building **production-grade ML/DL systems** from research to deployment. Leading on-device **phishing URL detection** using Transformer fine-tuning (MiniLM + LoRA) on **40M URLs**—achieving **98.91% AUC-ROC with 10% FNR, 1% FPR** and deployed as an Android app via ONNX INT8 quantization (**32.6 MB, <100ms inference**). Proficient in **Generative AI**: RAG pipelines, LLM fine-tuning, prompt engineering with LangChain, Pinecone, and ChromaDB. Experienced in end-to-end MLOps (Docker, FastAPI, Streamlit). Seeking an **AI/ML Engineer** or **Data Scientist** role at a product-driven organization.

## WORK EXPERIENCE

### AI Researcher — Cybersecurity AI Division

2025 – Present

IIT Ropar   **RESEARCH**

- Fine-tuned **MiniLM-L12-H384 Transformer + LoRA** (r=32, alpha=64) on **40M URLs Dataset** for phishing detection; achieved **AUC-ROC 98.91%**, FNR ≤10%, and **95.38% accuracy** on 4.4M-URL test set.
- Designed 6-stage canonical URL preprocessing pipeline handling Unicode homoglyphs, IDNA 2008 Punycode, recursive percent-decoding (10-pass), and high-entropy blob masking across **40M+ URLs**.
- Deployed production model as **PhishGuard Android App**—ONNX INT8 quantization reduced model to **32.6 MB** with **<100ms on-device inference**; fully offline, zero data exfiltration, validated via two-stage golden vector test.

PyTorch   Transformers   LoRA   ONNX Runtime   Focal Loss   Kotlin   Android   Jetpack Compose

### Data Scientist Intern

Apr 2025 – May 2025

AISPRY, Hyderabad, Telangana

- Built AI-powered **Autism Spectrum Detection system** using TensorFlow Lite + MediaPipe for pose estimation from video/webcam input; achieved **91.8% accuracy**.
- Integrated NLP-based multimodal analysis with a Gemini-powered chatbot for enhanced user interaction.
- Deployed scalable Flask + React application using Docker; automated data pipelines reducing preprocessing time by **15%**.

TensorFlow Lite   MediaPipe   NLP   React   Flask   Docker   Gemini API

### Data Science & Analytics Intern

Aug 2024 – Nov 2024

Zidio Development, Bengaluru, Karnataka

- Developed and deployed TensorFlow-based ML models in cross-functional teams, improving system efficiency by **15%**; automated data workflows with Python + SQL, boosting operational throughput by **25%**.
- Built interactive Power BI dashboards delivering actionable business insights to stakeholders.

Python   TensorFlow   Power BI   SQL

WORK EXPERIENCE (continued)

Data Science Trainee

Feb 2023 – Feb 2024

360DigiTMG, Hyderabad, Telangana

- Engineered predictive maintenance classification model with Scikit-learn; reduced machine downtime by **20%** through feature engineering on sensor data and failure pattern analysis.
- Developed behavior detection system for AI Tutor platform using classification models, increasing user engagement by **10%**; delivered Power BI dashboards for data-driven stakeholder insights.

Scikit-learn Python SQL Power BI

KEY PROJECTS

PhishGuard — On-Device Phishing URL Detection

github.com/SHINU4RATHOD/PhishDetect

★ AUC-ROC 98.91% · 35.3M URLs Trained · 32.6 MB ONNX INT8 · <100ms on-device inference

IIT Ropar Cybersecurity AI research: fine-tuned MiniLM-L12-H384 Transformer + LoRA ( $r=32$ ,  $\alpha=64$ ) on 35.3M URLs with Focal Loss and FP16 mixed precision, achieving **AUC-ROC 98.91%** and **95.38% accuracy** on a 4.4M-URL held-out test set. Built 6-stage URL canonicalization pipeline (NFKC normalization, IDNA 2008, recursive percent-decoding, high-entropy blob masking). Deployed as PhishGuard Android app via INT8 ONNX quantization — 32.6 MB, <100ms on-device inference, fully offline, zero data exfiltration.

PyTorch Transformers LoRA ONNX Runtime Focal Loss Kotlin Android NNAPI Jetpack Compose

AI-driven Autism Spectrum Detection (ASD)

GitHub ↗

★ 91.8% accuracy · End-to-end ML pipeline · Supports early ASD diagnosis for clinicians

End-to-end ML system for Autism Spectrum Disorder prediction from video/webcam input. Leveraged MediaPipe pose estimation and TensorFlow Lite binary classification, integrated with React frontend, Flask REST API, and a Gemini-powered conversational chatbot. Deployed via Docker for scalable clinical use.

TensorFlow Lite MediaPipe NLP React Flask Docker Gemini API

LLM-Powered RAG Applications (Medical Chatbot & Code Analyzer)

GitHub ↗

★ 30% faster retrieval · 25% code review efficiency gain · RAG with source attribution

Built two production RAG systems using LangChain, LLMs, and vector databases: (1) Medical chatbot with Pinecone for healthcare Q&A with source citations + Multi-PDF semantic search via FAISS & Gemini; (2) AI code analyzer using ChromaDB for intelligent codebase understanding, semantic search, and automated review recommendations.

LangChain LLMs RAG Pinecone ChromaDB FAISS Gemini Streamlit

Optimization of Machine Downtime — Predictive Maintenance

GitHub ↗

★ 20% reduction in unplanned downtime · Data-driven maintenance scheduling

Predictive maintenance classification pipeline using Scikit-learn to minimize industrial machine downtime. Applied feature engineering on sensor telemetry data to identify critical failure patterns; reduced unplanned downtime by 20%. Delivered real-time Power BI dashboards for operations stakeholders.

Scikit-learn Python SQL Power BI Feature Engineering

ADDITIONAL

Soft Skills: Problem-Solving Research & Analysis Cross-Functional Collaboration Technical Communication Agile / Scrum

Interests: Reading research papers & tech blogs · Open-source contributions · Fitness & running · Networking